

Multi-Label Learning for Reliable Cervical Cytology Screening

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Abstract. Cervical cytology screening requires automated and reliable assistance due to the labor-intensive nature and inter-observer variability of manual evaluation. Cropped tile classification models enable efficient analysis of gigapixel whole slide images but typically adopt binary or single-label formulations, which overlook coexisting lesion types and provide limited lesion localization, thus restricting clinical interpretability and diagnostic reliability. This study proposes MLCerNet, a Multi-Label Learning Cervical Screening Network that reformulates tile analysis as a multi-label classification task while producing lesion localization cues. To distinguish subtle morphological variations between lesion types, MLCerNet incorporates a Bilateral Attention Query (BiQuery) module that strengthens semantic interactions between lesion-specific and cell-wise features. To detect sparsely distributed lesions among abundant cells, it adopts a SimMat strategy that improves prediction robustness through a similarity matrix. By enabling lesion-specific CLS tokens to interact with fine-grained cell-wise tokens, the BiQuery module facilitates the modeling of subtle category-dependent morphological patterns. Building on these enhanced token representations, the SimMat strategy preserves category-aware token responses for prediction and provides localization cues under only tile-level label supervision. Extensive experiments demonstrate improved accuracy with enhanced interpretability for clinical practice. The code is available at <https://github.com/imAzhou/MLCerNet>.

Keywords: Cervical cytology screening, Multi-label classification.

1 Introduction

Cervical cancer is the fourth most common malignancy in women [25], and early diagnosis via cervical cytology is essential as the disease is largely preventable and treatable [11]. However, manual cytology slide screening is labor-intensive and prone to inter-observer variability, highlighting the need for computer vision models that provide automated, efficient, and accurate assistance [11]. Cropped tile classification models [4, 30, 24], which are prominent in automated screening, analyze gigapixel

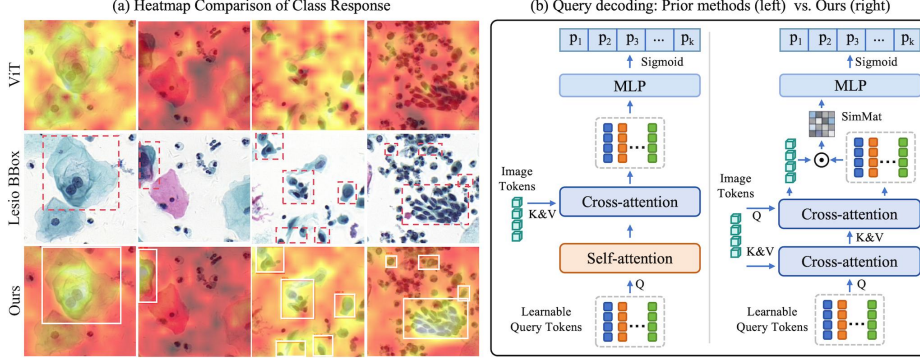


Fig. 1. (a) Compared to ViTs, our method exhibits stronger class responses when using only category labels (red dashed lines indicate the actual lesion locations). (b) Architectural comparison between the prior multi-label decoder and ours.

whole slide images (WSIs) by dividing them into smaller tiles. These models offer rapid inference and high accuracy in identifying positive slides, making them suitable for large-scale preliminary screening. However, existing cropped tile classification models still face two notable limitations. First, a single tile often contains multiple coexisting lesion types, whereas widely used models simplify the task into binary classification [4, 24] (Negative/Positive) or a single-label multi-class formulation [30], in which only the most severe lesion is preserved. Second, models such as those based on ViT rely solely on category labels as weak supervision and struggle to localize lesion regions, resulting in a black-box with limited interpretability and insufficient reliability for clinical practice, as shown in Fig. 1(a).

In this paper, we propose a Multi-Label Learning Cervical Screening Network (MLCerNet), a novel framework that reformulates tile analysis as a multi-label classification (MLC) task while enabling lesion localization, which in turn leads to improved interpretability and reliability. The MLC task estimates the probability of each lesion category being present within a tile. However, prior MLC methods, such as Q2L [17] and ML-Decoder [22], exhibit limited sensitivity when applied to cervical cytology screening, often resulting in missed positive lesions. The primary challenges arise from two aspects: first, the visual differences between certain lesion categories (e.g., ASC-US and LSIL) are subtle; second, lesions are often sparsely distributed within tiles that are predominantly composed of normal cells. To address these challenges, MLCerNet optimizes the multi-label decoding process of queries. As illustrated in Fig. 1(b), compared with prior methods, MLCerNet introduces two key modifications: (1) the conventional unidirectional cross-attention querying mechanism is reformulated, and (2) label probabilities are predicted using a similarity matrix rather than directly performing classification on query tokens.

Specifically, to capture inter-class feature discrepancies, a Bilateral Attention Query (BiQuery) module is designed as a bidirectional cross-attention mechanism between learnable lesion CLS query tokens and fine-grained cell-wise image tokens, as depicted in Fig. 2(b)(1). Moreover, to address the sparsity of lesion regions, a

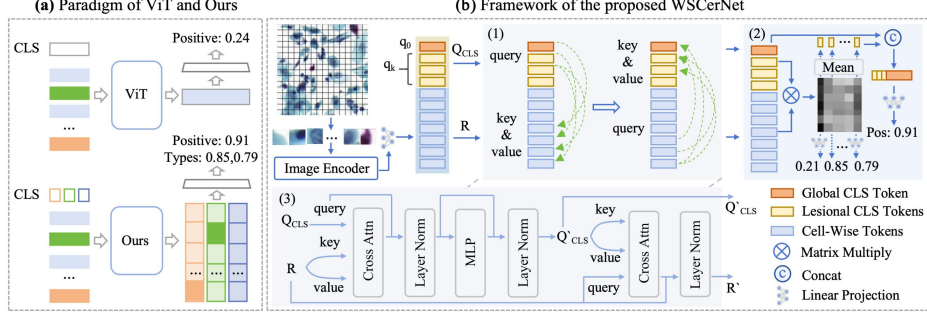


Fig. 2. (a) Classification paradigm comparison: ViT uses a single CLS token, whereas our method employs a similarity matrix between CLS and image tokens. (b) Framework of the proposed MLCerNet, including (1) bilateral attention query (BiQuery) module, (2) similarity-based classification mechanism, and (3) detailed structure of the BiQuery module.

classification strategy termed SimMat is introduced. It predicts the presence probability of each lesion category within a tile by leveraging a similarity matrix rather than compressing all cell-token information into a single CLS token before classification, as illustrated in Fig. 2(a). Extensive experiments on both public and private datasets demonstrate that MLCerNet achieves high sensitivity for positive tiles as well as strong accuracy in multi-label classification. Even with only category-level supervision, the model can effectively attend to lesion-related regions. These strengths collectively bridge the gap between black-box predictions and clinical interpretability. The main contributions of this work are summarized as follows:

- (1) We reformulate cervical cytology tile analysis as a multi-label classification task and propose MLCerNet to better reflect the co-existence of multiple lesion categories in practical screening scenarios.
- (2) A bilateral attention query (BiQuery) module is designed to enhance interactions between lesion-specific CLS tokens and fine-grained cell-wise tokens.
- (3) A similarity-matrix-based classification strategy is introduced to predict lesion-category probabilities while supporting interpretable lesion localization.

2 Related Work

Cervical Cytology Screening. Driven by the imperative for automated cervical cytology screening, numerous studies have explored computer vision-assisted methods [11, 12, 28]. Based on different task designs [11], these methods can be broadly categorized into tile classification [4, 24, 30] and abnormal cell detection [3, 7, 15]. Among these, tile classification models have emerged as a promising direction for the preliminary screening of cytology WSIs, owing to their simplified preprocessing requirements and rapid inference speeds. Typically, a fixed-size sliding window is used to crop square tiles from WSIs, non-cellular tiles are discarded, and the remaining tiles are processed by a classification model to extract 1D feature vectors for downstream WSI-level prediction [11]. However, existing tile

classification methods often oversimplify the task into binary classification (Negative/Positive) [4, 24] or a single-label multi-class formulation where only the most severe lesion is preserved [30]. This paradigm is inconsistent with clinical reality, as multiple distinct lesion types frequently co-occur within a single tile. Consequently, this discrepancy underscores the need for a more comprehensive label assignment strategy, motivating the shift toward multi-label learning.

Multi-Label Classification. To address the limitations of single-label paradigms, Multi-Label Classification (MLC) presents a promising alternative, as it allows for the simultaneous recognition of multiple lesion categories within a single view [18]. Before the introduction of the transformer attention mechanism [26], multi-label classification in deep learning primarily relied on CNNs to extract image features, followed by convolutional feature aggregation to predict the presence probabilities of each category [21, 31]. With the advent of transformers, subsequent studies began adopting attention-based mechanisms for multi-label prediction [14, 17, 22]. Due to its capability to capture complex semantic relationships, MLC has been increasingly applied to various medical image classification tasks [5, 19, 29]. Despite its potential, MLC remains underexplored in cervical cytology tile analysis due to subtle inter-class visual differences and sparsely distributed lesions within predominantly normal-cell regions. Consequently, general-purpose MLC methods struggle to capture fine-grained and localized features, resulting in suboptimal performance in this domain.

3 Method

In this section, the proposed MLCerNet is detailed, which performs multi-label classification based on a similarity matrix between CLS tokens and image tokens (Fig. 2(a)). The proposed Bilateral Attention Query module is detailed in Section 3.1, and the SimMat strategy is described in Section 3.2.

3.1 Bilateral Attention Query

To enhance semantic correspondences between CLS tokens and image tokens, we propose a Bilateral Attention Query (BiQuery) module, which is inspired by the location prompt-based attention mechanism in SAM [13]. In our design, the location prompts are replaced with class embedding queries, including category-specific Lesion CLS tokens and a global Positive CLS token, enabling class-aware attention responses from cell-wise image tokens.

First, a cell-wise token division mechanism is designed to extract fine-grained cellular features. Given an input tile I , the image is divided into $T = n \times n$ tokens, each encoded as a feature vector of dimension $1 \times d$. Considering that the average cell size is $(h_{\text{cell}}, w_{\text{cell}})$, the tile cropped from the WSI is set to $(n \times h_{\text{cell}}, n \times w_{\text{cell}})$ to ensure that each token approximately corresponds to a single cell. It is important to note that $(h_{\text{cell}}, w_{\text{cell}})$ varies with scanning magnification (e.g., $20\times$ or $40\times$) and cellular types (e.g., larger superficial cells vs. smaller basal cells). Thus, a statistical estimate (e.g., median or mean) is adopted to determine their values. After

embedding the image into cell-wise tokens, the token sequence is fed into the transformer-based encoder to extract cellular representations, where self-attention enables interactions among cell-wise tokens even when some tokens partially contain cells due to uniform partitioning. To address variations in feature dimensionality across different backbones, a linear projection layer is applied to map the encoded features into a unified token space, producing the final cell-wise representation set

$$R = \{r_t\}_{t=1}^T, \text{ where } r_t \in \mathbb{R}^{1 \times d}.$$

Then, a global CLS token q_0 and a set of lesion-specific CLS tokens q_k ($k = 1, 2, \dots, K$) are defined, where q_0 is responsible for determining whether R contains abnormal cells, while each q_k estimates the probability of the corresponding lesion type being present. Here, K denotes the number of predefined lesion categories. These learnable CLS tokens, $\{q_k\}_{k=0}^K$ (collectively denoted as Q_{CLS}), are then fed into the BiQuery module together with the cell-wise tokens R , as illustrated in Fig. 2(b)(2). In cytology analysis, determining whether a cell is abnormal and identifying its lesion type requires comparison with neighboring cells, as examining a single cell in isolation is insufficient. Therefore, each q_k in BiQuery computes attention with every token r_t , and subsequently, each r_t computes attention with every q_k . This process is expressed as:

$$Q'_{\text{CLS}} = \mathcal{F}_{\text{attn}}^1(Q_{\text{CLS}}, R), R' = \mathcal{F}_{\text{attn}}^2(R, Q'_{\text{CLS}}), \quad (1)$$

where $\mathcal{F}_{\text{attn}}^1$ and $\mathcal{F}_{\text{attn}}^2$ represent two different groups of attention modules. The details of the BiQuery module are shown in Fig. 2(b)(3). The input Q_{CLS} , R and the output Q'_{CLS} , R' are dimensionally consistent, allowing the BiQuery module to be stacked. The number of stacking layers, M , is a tunable hyper-parameter.

3.2 Classification via Similarity Matrix

Conventional transformer-based classification methods typically aggregate information from all image tokens into a single CLS token, which is then linearly projected for predicting the presence probabilities of different categories. In contrast, to robustly discriminate lesions from abundant normal-cell backgrounds, we design a classification strategy termed SimMat, which performs multi-label prediction by directly leveraging token–category similarity measurements rather than compressing cellular information into one CLS representation. As illustrated in Fig. 2(b)(2), the updated lesional CLS tokens Q'_{CLS} and the updated cell-wise tokens R' from the BiQuery module are multiplied to form a similarity matrix, referred to as SimMat. Here, each column of SimMat represents the attention intensity of a CLS token toward each cell-wise token r_t . A linear projection is applied to each column to obtain the presence probability p_k for the k -th lesion category. Specifically, the linear projection performs a weighted summation over each column, followed by a sigmoid function σ , formulated as:

$$p_k = \sigma(W_k(q_k R^T) + b_k), k \in \{1, 2, \dots, K\}, \quad (2)$$

where T denotes matrix transposition. Meanwhile, the mean attention intensity across each column of SimMat is concatenated with the updated global CLS token q'_0 , resulting in a joint representation $q_{\text{global}} \in \mathbb{R}^{1 \times (d+K)}$. This representation is subsequently passed through a linear projection layer to obtain the positive probability p_0 , which indicates the existence of lesion regions within the tile.

With the obtained probabilities p_k and p_0 , the binary cross-entropy (BCE) loss $\mathcal{L}$ is calculated with GT label y_k :

$$\mathcal{L} = \sum_{k=0}^K \text{BCE}(p_k, y_k), k \in \{0, 1, 2, \dots, K\}. \quad (3)$$

Then, the classification loss $\mathcal{L}$ serves as the supervision signal to update the model parameters. The attention of each lesional CLS token progressively concentrates on the most relevant cell-wise tokens of its corresponding category. As a result, the model can localize potential lesion regions using only category labels, thereby improving both transparency and reliability.

4 Experiments

Datasets. Experiments were conducted on both a public dataset, *CDetector* [16], and a privately collected dataset, the Large-scale Cervical Screening (*L-CerScan*) dataset. The *CDetector* dataset was collected for abnormal cervical lesion detection and contains 6,666 training images and 744 test images with varying image sizes. We crop square tiles from these images and assign the tile's category GT label based on the abnormal bounding boxes within the tile.

To further validate MLCerNet for cervical cytology screening, 1,021 positive cervical cytology slides from *L-CerScan* were collected and divided into training and test sets with an 8:2 ratio. Each cropped tile was independently annotated by two experienced pathologists according to the lesion types visible within the image, while tiles without any lesions were labeled as negative. Both datasets rely solely on category labels for model training. Square tiles for both datasets follow the design principles in Section 3.1, with *CDetector* cropped to 400×400 px and *L-CerScan* to 700×700 px. The distribution of cropped tiles is shown in Fig. 3.

Implementation Details. In MLCerNet, ViT-L serves as the encoder for feature extraction. The images are resized to 224×224 px and partitioned into 196 cell-wise tokens using a 16×16 sliding window. These tokens are then processed by an encoder initialized with pre-trained weights from the cytology model SmartCCS [10]. The number of BiQuery layers M is set to 2. All comparison methods use the AdamW optimizer and are trained for 100 epochs with a batch size of 128 on 8 A800 GPUs.

Metrics. Accuracy (Acc.), sensitivity (Sens.), and specificity (Spec.) are used to evaluate binary image classification (positive vs. negative), while the F1 score assesses the performance of each positive category.

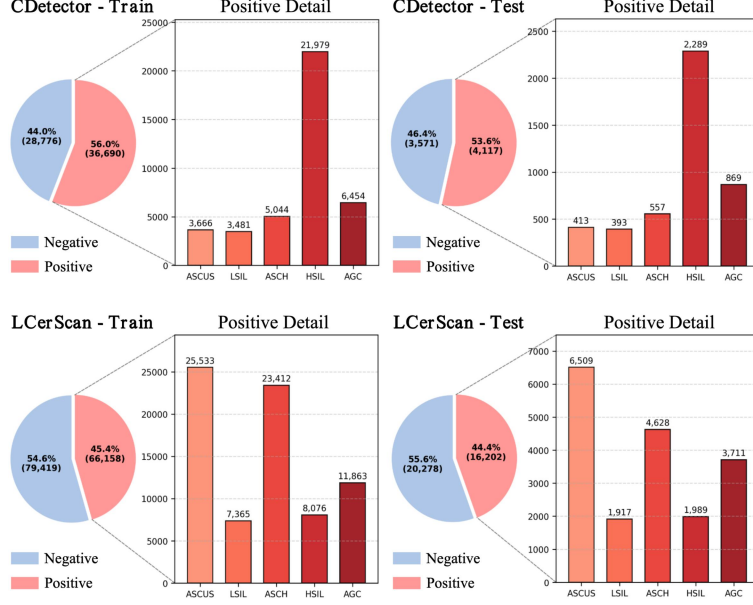


Fig. 3. Negative vs. positive tiles and positive tile distribution in *CDetector* and *L-CerScan* (train/test). A single tile may contain multiple lesion types.

4.1 Result Analysis

Lesion classification compared with SOTA methods. The proposed MLCerNet was evaluated against state-of-the-art classification models on both *CDetector* and *L-CerScan*. For ResNet50 [9], the spatial feature maps are processed by global average pooling to obtain a 1D feature vector. For ViT-based architectures, including ViT-L [6], DINOv2-L [20], UNI [2], and SmartCCS [10], the CLS token is used as the 1D feature representation, where it is updated through attention interactions with image tokens. Subsequently, the resulting 1D feature vector is projected into lesion-category presence probabilities through a linear classification layer. As shown in Table 1, SmartCCS achieves the best performance due to its pre-training on large-scale cytology tiles in an unsupervised manner. Therefore, SmartCCS is adopted as the backbone feature extractor for all subsequent classifiers. To further validate the effectiveness of MLCerNet as a multi-label classifier, we additionally evaluated three representative multi-label methods, Q2L [17], ML-Dec [22] and CIPL [27], as well as the lesion-specific model HierSwin [1]. Here, HierSwin supports only binary classification. As shown in Table 1, MLCerNet surpasses all baselines in binary classification and positive-class recognition, achieving nearly 3% higher sensitivity, which underscores its superior capability in positive sample detection. To assess performance on small lesions, a subset of *CDetector* was extracted containing at least one lesion smaller than 64x64 px per image. UNI, Q2L, and ML-Decoder were evaluated on this subset. Table 2 shows our method achieves the highest sensitivity, highlighting its effectiveness on small lesions.

Table 1. Comparison results on *CDetector* and *L-CerScan*. Bold indicates the best, underlined the second best. F1 score is used for each positive category.

	Methods	Year	Acc.	Sens.	Spec.	ASCUS	LSIL	ASCH	HSIL	AGC
<i>CDetector</i>	ResNet50 [9]	2016	77.64	71.99	84.15	44.77	46.65	24.88	76.03	70.30
	ViT-L [6]	2021	84.83	84.96	84.68	54.02	<u>63.77</u>	37.97	83.38	78.46
	DINOv2 [21]	2023	85.60	84.21	85.20	<u>59.26</u>	62.95	<u>40.13</u>	84.16	78.82
	UNI [2]	2024	85.18	83.26	84.78	58.14	63.11	35.16	83.98	77.92
	SmartCCS [10]	2025	85.67	85.55	85.81	59.12	62.99	39.87	83.76	78.32
	Q2L [17]	2021	80.88	77.29	85.02	46.19	51.51	22.66	81.36	72.01
	ML-Decoder [24]	2023	84.87	<u>85.87</u>	84.18	58.25	61.91	36.72	<u>85.12</u>	<u>78.85</u>
	HierSwin [1]	2024	<u>85.77</u>	84.33	<u>86.27</u>	-	-	-	-	-
	CIPL [30]	2025	83.28	82.96	83.65	49.04	50.83	32.77	84.97	76.80
	Ours	-	87.55	88.34	86.64	59.40	63.98	45.32	85.28	80.16
<i>L-CerScan</i>	ResNet50 [9]	2016	87.63	82.55	91.68	56.05	26.31	53.37	37.02	55.23
	ViT-L [6]	2021	88.46	81.69	93.88	61.61	34.34	52.63	36.73	56.07
	DINOv2 [21]	2023	<u>93.28</u>	91.77	94.49	66.34	39.59	59.61	39.84	61.32
	UNI [2]	2024	93.01	91.03	<u>94.59</u>	66.45	38.86	58.62	41.03	58.43
	SmartCCS [10]	2025	93.45	92.44	94.26	67.05	42.98	58.99	43.63	<u>65.58</u>
	Q2L [17]	2021	92.16	89.67	94.15	66.97	42.55	59.64	43.33	63.15
	ML-Decoder [24]	2023	93.10	91.34	94.50	<u>67.43</u>	42.62	<u>59.82</u>	42.82	64.63
	HierSwin [1]	2024	93.16	<u>92.48</u>	94.01	-	-	-	-	-
	CIPL [30]	2025	92.70	91.47	93.68	67.01	<u>43.59</u>	59.10	<u>44.86</u>	63.99
	Ours	-	95.21	95.43	95.03	68.77	45.81	62.91	47.89	67.12

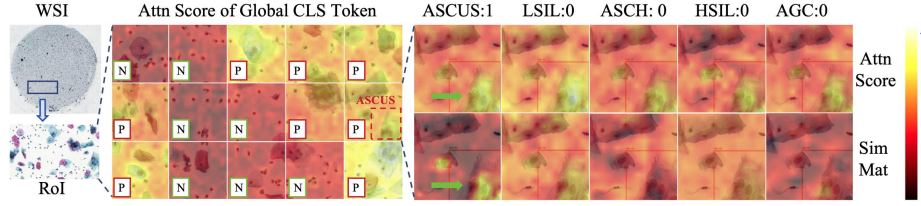


Fig. 4. Visualization of attention scores (AttnScores) and similarity matrix (SimMat). The global CLS token highlights positive tiles (denoted as "P") within an ROI. Although all lesion CLS tokens achieve high AttnScores (indicated by green arrows), only the correct token exhibits the highest similarity in SimMat.

Lesion localization performance analysis. First, we present the localization visualization results of MLCerNet on regions of interest (ROI) and tiles in cytology WSIs. The CLS tokens, including global and lesion tokens, compute attention scores and similarity values with the cell-wise tokens. The resulting AttnScores and SimMat are shown in Fig. 4. As observed, for all tiles within a RoI, the global CLS token exhibits stronger attention on positive tiles. For a selected positive tile, although all lesion CLS tokens attend to the lesion region (indicated by green arrows), only the CLS token of the correct category (ASC-US in this example) exhibits the highest similarity in SimMat. Qualitatively, the SimMat effectively capture the correspondence between categories and regions. Subsequently, object detection performance was evaluated using bounding bboxes obtained from SimMat post-processing. Following [8], heatmap regions in SimMat were converted into bounding

Table 2. Model performance on *CDetector*: (1) classification on the small-lesion dataset and (2) localization comparison with detectors.

Classification on small-lesion					Location on cropped tiles			
Metric	UNI	Q2L	ML-Dec	Ours	Metric	DINO	YOLOv12	Ours
Acc.	90.66	90.13	90.95	92.47	mAP	47.28	52.13	35.76
Sens.	85.21	86.06	86.49	89.86	mAR	70.85	67.28	51.01
Spec.	94.83	93.42	94.35	95.30	FPS	20.14	82.76	120.96

bboxes, each of which was assigned to the category of the lesion token whose area above a specified similarity threshold in SimMat was largest. The localization results are summarized in Table 2, where mAP and mAR denote the mean precision and recall of the predictions, respectively, and FPS indicates the number of tiles processed per second by the model. These results indicate that MLCerNet achieves competitive localization performance under weak supervision with only class labels, while fully supervised detectors still retain an advantage. However, in terms of inference speed (FPS), MLCerNet demonstrates the efficiency advantage of a classification model.

4.2 Ablation Study

Ablation studies were conducted on *L-CerScan*, examining the cell-wise token division mechanism ("CWTD"), the bilateral attention query module ("BiQuery"), and the similarity matrix classification ("SimMat"). CWTD was ablated by reducing the cropping size from 700 px to 392 px, BiQuery by setting $M = 0$, and SimMat by performing classification directly on Q_{CLS} . Each component was ablated independently while keeping the others unchanged. The results in Table 3 demonstrate the contribution of each component. Removing CWTD leads to a clear performance drop, indicating that cell-wise token division is beneficial for preserving fine-grained cellular morphology. Without BiQuery, the model shows reduced sensitivity, suggesting that bidirectional interactions between lesion-specific CLS tokens and cell-wise tokens are important for recognizing subtle abnormal patterns. The most significant degradation is observed when SimMat is removed, confirming that similarity-based prediction is more effective than directly compressing tile information into CLS representations. Additionally, an ablation on M yielded accuracy scores of 93.98, 94.81, 94.13, and 94.01 for $M = 1, 2, 3$, and 4, respectively, supporting the choice of $M = 2$ for MLCerNet.

In addition, we trained MLCerNet with three tile-level objectives and then used it as a feature extractor for WSI classification: (i) binary classification of negative versus positive tiles, (ii) multi-class classification based on the most severe lesion, and (iii) the proposed multi-label classification (MLC). Tile features extracted from the *L-CerScan* WSI dataset were fed into TransMIL [23] for WSI-level prediction. It is noted that *L-CerScan* contains only 1,021 positive WSIs, as detailed in Section 4. We additionally collected 1,100 negative WSIs to conduct a WSI binary classification experiment. The results are presented in Table 3, showing that using MLC as the

Table 3. Ablation studies on *L-CerScan*: (1) effect of three key components at the tile-level, and (2) effect of training objectives at the WSI-level.

Metric	Ablation of key components				Ablation of training objectives		
	-CWTD	-BiQuery	-SimMat	Ours	Binary	Severest	MLC
Acc.	93.82	93.94	92.17	95.21	85.11	86.30	89.28
Sens.	93.26	95.39	92.95	95.43	84.02	<u>85.12</u>	90.24
Spec.	94.27	92.81	91.55	95.03	86.12	87.03	88.30

training objective clearly helps the feature extractor capture finer-grained features, thereby enhancing its performance in WSI classification. Compared with binary or most-severe-lesion supervision, MLC preserves category-aware information from coexisting lesions within each tile. These richer tile-level representations improve the discriminative ability of downstream WSI-level classification.

5 Conclusion

In this paper, we presented MLCerNet, a multi-label learning framework tailored for cervical cytology screening that enhances both diagnostic accuracy and interpretability. By explicitly modeling the coexistence of multiple lesion types and strengthening semantic interactions between lesion-specific and cell-wise tokens through the BiQuery module, MLCerNet effectively distinguishes subtle morphological variations. In addition, the proposed SimMat classification strategy enables transparent and reliable lesion localization, effectively identifies sparsely distributed lesions among abundant normal cells, and mitigates the black-box limitations of conventional tile-based approaches. Comprehensive evaluations on the public *CDetector* and private *L-CerScan* datasets show that MLCerNet improves sensitivity and delivers competitive multi-label classification performance. However, multi-label learning for cervical cytology still has considerable room for advancement. For instance, MLCerNet lacks probability constraints, which may lead the model to predict a tile as negative while assigning a high probability to lesion presence, which is inconsistent with clinical reality. Moreover, lesion pairs that often co-occur clinically such as ASC-US with LSIL or ASC-H with HSIL may lead the model to assign similarly high probabilities to both categories even when only one is present, reducing its ability to distinguish closely related lesions. Future work may incorporate clinically informed relationships through hierarchical or structure-aware losses to better couple positive and category specific predictions, leading to a more robust and clinically faithful multi-label formulation.

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Disclosure of Interests. The authors have no competing interests to declare.

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